Molecular Systems in Chemical Engineering –Spring 2022

CHE 230 Self-Study Report
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Course Comments / Assessment Summary

Assessment of student performance indicates variation across performance indicators. In particular, students demonstrated the greatest difficulty with problems requiring multi-step interpretation and sustained calculation (PI 1/2), while showing stronger performance in identifying appropriate analytical methods (PI 4/6).

For PI 1/2, many students struggled to interpret the problem statement and determine the required steps. A significant number of students did not attempt the question. Feedback collected after the assessment suggests that, although students recognized the problem as solvable, the level of effort required was not aligned with its relatively low point value. This may have contributed to reduced motivation to attempt the problem.

For PI 4/6, students who attempted the problem were generally able to correctly identify the relevant equations, indicating competency in selecting appropriate analytical tools. However, several students did not complete the solution process, stopping after listing equations without substituting values or performing calculations. This suggests a gap between conceptual recognition and full execution of problem-solving procedures.

Course Reflections

Some students continue to demonstrate challenges with foundational academic skills, including careful reading of problem statements, structured problem-solving, and persistence on multi-step questions. Additionally, there appears to be a tendency among some students to disengage from problems perceived as time-consuming or insufficiently weighted.

These observations are consistent with broader concerns regarding student study habits, including preparation before class, engagement during instruction, and follow-through on assignments.

Planned Improvements

- Assessment Design:

Point values will be revised to better reflect the level of effort, complexity, and time required for each question. Multi-step problems will carry increased weight to encourage student engagement.

- Problem-Solving Emphasis:

Instruction will place greater emphasis on complete problem-solving processes, including interpretation, equation selection, numerical substitution, and solution verification.

- Student Support Resources:

Students will be permitted to use a structured cheat sheet, and an equation sheet will be provided to reduce cognitive load associated with memorization and allow greater focus on application.

- Early Feedback:

Additional low-stakes assessments (e.g., quizzes) will be introduced earlier in the semester to provide timely feedback and encourage consistent engagement.

- Active Learning Strategies:

Class time will incorporate more guided problem-solving activities to help students practice interpreting and solving multi-step problems in a supported environment.